

FRACTIONS - SHORT REVISION NOTES

Create Level: Produce Something New

1. Creating Fractions from Given Conditions

Condition	Fraction Created	Explanation
Numerator = 3, Denominator = 5	$\frac{3}{5}$	3 parts out of 5 equal parts
Numerator = 7, Denominator = 8	$\frac{7}{8}$	7 parts out of 8 equal parts
Numerator = 5, Denominator = 12	$\frac{5}{12}$	5 parts out of 12 equal parts
A fraction greater than 1	$\frac{7}{4}$ or $1\frac{3}{4}$	Improper fraction
A fraction between 0 and 1	$\frac{2}{3}$ or $\frac{3}{4}$	Proper fraction
A unit fraction	$\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$	Numerator = 1

Create Your Own Fractions:

Task	Your Creation
Create a proper fraction	_____
Create an improper fraction	_____
Create a mixed number	_____
Create a unit fraction	_____
Create two equivalent fractions	_____

2. Creating Equivalent Fractions

Method: Multiply or divide the numerator and denominator by the SAME number.

Original Fraction	Created Equivalent Fractions
$\frac{1}{2}$	$\frac{2}{4}$, $\frac{3}{6}$, $\frac{4}{8}$, $\frac{5}{10}$, $\frac{6}{12}$
$\frac{1}{3}$	$\frac{2}{6}$, $\frac{3}{9}$, $\frac{4}{12}$, $\frac{5}{15}$, $\frac{6}{18}$
$\frac{3}{4}$	$\frac{6}{8}$, $\frac{9}{12}$, $\frac{12}{16}$, $\frac{15}{20}$
$\frac{2}{5}$	$\frac{4}{10}$, $\frac{6}{15}$, $\frac{8}{20}$, $\frac{10}{25}$
$\frac{4}{5}$	$\frac{8}{10}$, $\frac{12}{15}$, $\frac{16}{20}$, $\frac{20}{25}$

Create Equivalent Fraction Pairs:

Original	Multiply by	Equivalent Fraction
$\frac{2}{3}$	x2	$\frac{4}{6}$
$\frac{2}{3}$	x3	$\frac{6}{9}$
$\frac{2}{3}$	x4	$\frac{8}{12}$
$\frac{2}{3}$	x5	$\frac{10}{15}$

3. Creating Mixed Numbers and Improper Fractions

From a Given Mixed Number:

Mixed Number	Convert to Improper	Check Your Work
2 $\frac{3}{4}$	$\frac{11}{4}$	$(2 \times 4 + 3 = 11)$ ✓
3 $\frac{2}{5}$	$\frac{17}{5}$	$(3 \times 5 + 2 = 17)$ ✓
1 $\frac{7}{8}$	$\frac{15}{8}$	$(1 \times 8 + 7 = 15)$ ✓
4 $\frac{1}{3}$	$\frac{13}{3}$	$(4 \times 3 + 1 = 13)$ ✓

From a Given Improper Fraction:

Improper	Convert to Mixed	Check Your Work
$\frac{7}{3}$	2 $\frac{1}{3}$	$7 \div 3 = 2$ R1 ✓
$\frac{11}{4}$	2 $\frac{3}{4}$	$11 \div 4 = 2$ R3 ✓
$\frac{27}{8}$	3 $\frac{3}{8}$	$27 \div 8 = 3$ R3 ✓
$\frac{36}{7}$	5 $\frac{1}{7}$	$36 \div 7 = 5$ R1 ✓

Create Your Own Conversions:

Task	Your Conversion
Convert 3 $\frac{1}{4}$ to improper	_____
Convert $\frac{19}{5}$ to mixed	_____
Create an improper fraction and convert to mixed	_____
Create a mixed number and convert to improper	_____

4. Creating Fraction Addition Problems

Condition	Created Problem	Answer
Same denominator	$\frac{3}{5} + \frac{1}{5}$	$\frac{4}{5}$
Same denominator (different numerators)	$\frac{5}{6} + \frac{5}{6}$	$\frac{10}{6} = 1 \frac{2}{3}$
Different denominators	$\frac{1}{2} + \frac{1}{3}$	$\frac{5}{6}$
Mixed numbers	$1 \frac{1}{2} + 2 \frac{1}{4}$	$3 \frac{3}{4}$

Create Your Own Addition Problems:

Type	Your Problem	Answer
Same denominator	_____	_____
Different denominators	_____	_____
Mixed numbers	_____	_____

Create 3 Steps to Solve $\frac{1}{2} + \frac{1}{3}$:

Step 1: Find LCM of 2 and 3 = 6

Step 2: Convert: $\frac{1}{2} = \frac{3}{6}$, $\frac{1}{3} = \frac{2}{6}$

Step 3: Add: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

5. Creating Fraction Subtraction Problems

Condition	Created Problem	Answer
Same denominator	$4/5 - 1/5$	$3/5$
Different denominators	$3/4 - 1/2$	$1/4$
Mixed numbers	$3 \frac{1}{2} - 1 \frac{3}{4}$	$1 \frac{3}{4}$
With borrowing	$3 \frac{1}{5} - 1 \frac{3}{4}$	$1 \frac{9}{20}$

Create Your Own Subtraction Problems:

Type	Your Problem	Answer
Same denominator	_____	_____
Different denominators	_____	_____
Mixed numbers	_____	_____

Create 3 Steps to Solve $3/4 - 1/2$:

Step 1: Find LCM of 4 and 2 = 4

Step 2: Convert: $3/4 = 3/4$, $1/2 = 2/4$

Step 3: Subtract: $3/4 - 2/4 = 1/4$

6. Creating Fraction Multiplication Problems

Condition	Created Problem	Answer
Proper fractions	$2/3 \times 3/4$	$1/2$
Improper fractions	$5/3 \times 7/4$	$35/12 = 2 \frac{11}{12}$
Mixed numbers	$1 \frac{1}{2} \times 2 \frac{3}{4}$	$4 \frac{1}{8}$
With cancellation	$3/4 \times 8/9$	$2/3$

Create Your Own Multiplication Problems:

Type	Your Problem	Answer
Proper fractions	_____	_____
Mixed numbers	_____	_____
With cancellation	_____	_____

Create 3 Steps to Solve $2/3 \times 3/4$:

Step 1: Multiply numerators: $2 \times 3 = 6$

Step 2: Multiply denominators: $3 \times 4 = 12$

Step 3: Simplify: $6/12 = 1/2$

Create 3 Steps with Cancellation:

Step 1: $\frac{3}{4} \times \frac{8}{9}$

Step 2: Cancel: 3 with 9 ($\div 3$), 4 with 8 ($\div 4$)

Step 3: Multiply: $\frac{1}{3} \times \frac{2}{1} = \frac{2}{3}$

7. Creating Fraction Division Problems

Condition	Created Problem	Answer
Proper fractions	$2/3 \div 3/4$	$8/9$
Mixed numbers	$1\ 1/2 \div 2\ 1/4$	$2/3$
Whole $\div$ Fraction	$6 \div 3/4$	8
Fraction $\div$ Whole	$3/4 \div 3$	$1/4$

Create Your Own Division Problems:

Type	Your Problem	Answer
Proper fractions	_____	_____
Mixed numbers	_____	_____
Whole $\div$ Fraction	_____	_____

Create 3 Steps to Solve $2/3 \div 3/4$:

Step 1: Keep the first fraction: $2/3$
 Step 2: Change $\div$ to $\times$
 Step 3: Flip the second fraction: $4/3$
 Step 4: Multiply: $2/3 \times 4/3 = 8/9$

8. Creating "Of" Problems

Condition	Created Problem	Answer
Fraction of whole number	$3/4$ of 64	48
Fraction of fraction	$2/3$ of $1/5$	$2/15$
Mixed number of whole	$1\ 1/2$ of 20	30
Fraction of mixed	$4/5$ of $2\ 1/2$	2

Create Your Own "Of" Problems:

Type	Your Problem	Answer
Fraction of whole	_____	_____
Fraction of fraction	_____	_____
Mixed of whole	_____	_____

Method to Solve $3/4$ of 64:

Step 1: Replace "of" with $\times$: $3/4 \times 64$
 Step 2: $64 \div 4 = 16$
 Step 3: $16 \times 3 = 48$

9. Creating BODMAS Problems

Level	Created Problem	Answer
Easy	$1/2 + 1/4 \times 2/3$	$5/6$
Medium	$(1/2 + 1/4) \times 2/3$	$1/2$
Hard	$2/3 \text{ of } 3/4 + 1/2$	1
Complex	$1 \frac{1}{2} + 2 \frac{3}{4} \div 1/2$	7

Create Your Own BODMAS Problems:

Level	Your Problem	Answer
Easy	_____	_____
Medium	_____	_____
Hard	_____	_____

Create 3 Steps to Solve $1/2 + 1/3 \times 3/4$:

Step 1: Multiplication first: $1/3 \times 3/4 = 3/12 = 1/4$

Step 2: Addition: $1/2 + 1/4$

Step 3: Find LCM: $2/4 + 1/4 = 3/4$

10. Creating Reciprocal Problems

Number	Reciprocal Created	Check (Product = 1)
$2/3$	$3/2$	$2/3 \times 3/2 = 1$
$3/4$	$4/3$	$3/4 \times 4/3 = 1$
5	$1/5$	$5 \times 1/5 = 1$
$2 \frac{1}{2} (5/2)$	$2/5$	$5/2 \times 2/5 = 1$

Create Your Own Reciprocals:

Number	Your Reciprocal	Check
$3/5$	_____	_____
$7/8$	_____	_____
6	_____	_____
$3 \frac{1}{4}$	_____	_____

11. Creating Word Problems

Template 1: Money Problems

Created Problem	Solution
A person has Rs 100. He spends $\frac{1}{4}$ on food and $\frac{1}{2}$ on transport. How much does he save?	$100 - (25 + 50) = \text{Rs } 25$
A book costs Rs 80. If there is a discount of $\frac{1}{5}$, what is the price?	$80 \times \frac{1}{5} = 16$; $80 - 16 = \text{Rs } 64$

Create Your Own Money Problem:

Problem: _____

Solution: _____

Answer: _____

Template 2: Land/Area Problems

Created Problem	Solution
A farmer owns $\frac{3}{4}$ of a 20-hectare land. He plants $\frac{2}{3}$ of his land with rice. What area is planted with rice?	$\frac{3}{4} \times 20 = 15$ hectares; $\frac{2}{3} \times 15 = 10$ hectares
A person gives $\frac{1}{2}$ of his land to son and $\frac{1}{3}$ to daughter. What fraction is given away?	$\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$

Create Your Own Land Problem:

Problem: _____

Solution: _____

Answer: _____

Template 3: Journey Problems

Created Problem	Solution
Sam walked $\frac{1}{5}$ of a 15 km journey. How far did he walk?	$\frac{1}{5} \times 15 = 3$ km
A bus travels $\frac{3}{4}$ of a 40 km journey. What distance remains?	$\frac{3}{4} \times 40 = 30$ km; $40 - 30 = 10$ km

Create Your Own Journey Problem:

Problem: _____

Solution: _____

Answer: _____

Template 4: Sharing Problems

Created Problem	Solution
6 friends share $\frac{3}{4}$ of a cake. How much cake does each get?	$\frac{3}{4} \div 6 = \frac{3}{4} \times \frac{1}{6} = \frac{3}{24} = \frac{1}{8}$ of the cake
A pizza is cut into 8 slices. 3 friends eat $\frac{5}{6}$ of the pizza equally. How much does each eat?	$\frac{5}{6} \div 3 = \frac{5}{6} \times \frac{1}{3} = \frac{5}{18}$ of the pizza

Create Your Own Sharing Problem:

Problem: _____

Solution: _____

Answer: _____

12. Creating Worksheets

Create Your Own Worksheet: Fraction Addition

Problem	Answer Space
$\frac{2}{5} + \frac{1}{5} =$	_____
$\frac{3}{4} + \frac{1}{4} =$	_____
$\frac{1}{2} + \frac{1}{3} =$	_____
$1\frac{1}{2} + 2\frac{1}{4} =$	_____
$2\frac{1}{3} + 1\frac{2}{3} =$	_____

Create Your Own Worksheet: Fraction Multiplication

Problem	Answer Space
$\frac{1}{2} \times \frac{1}{3} =$	_____
$\frac{2}{3} \times \frac{3}{4} =$	_____
$1\frac{1}{2} \times 2\frac{1}{4} =$	_____
$\frac{3}{4} \times \frac{8}{9} =$	_____
$2\frac{1}{3} \times 1\frac{1}{2} =$	_____

Create Your Own Worksheet: Fraction Division

Problem	Answer Space
$\frac{1}{2} \div \frac{1}{3} =$	_____
$\frac{3}{4} \div \frac{1}{2} =$	_____
$1\frac{1}{2} \div 2\frac{1}{4} =$	_____
$6 \div \frac{3}{4} =$	_____
$\frac{2}{3} \div \frac{4}{5} =$	_____

13. Creating BODMAS Worksheet

Problem	Your Solution
$1/2 + 1/3 \times 1/4$	_____
$(1/2 + 1/3) \times 1/4$	_____
$2/3 \text{ of } 3/4 + 1/2$	_____
$1 \frac{1}{2} + 2 \frac{3}{4} \div 1/2$	_____
$(1 \frac{1}{2} + 2 \frac{3}{4}) \div 1/2$	_____

Create BODMAS Rule Summary:

B = Brackets: Example $\rightarrow$ _____

O = Of: Example $\rightarrow$ _____

D = Division: Example $\rightarrow$ _____

M = Multiplication: Example $\rightarrow$ _____

A = Addition: Example $\rightarrow$ _____

S = Subtraction: Example $\rightarrow$ _____

14. Creating Fraction Patterns

Pattern 1: Adding Same Denominator

1st Fraction	2nd Fraction	Sum
$1/5$	$1/5$	$2/5$
$1/5$	$2/5$	$3/5$
$1/5$	$3/5$	$4/5$
$1/5$	$4/5$	1

Create Your Own Pattern:

_____ + _____ = _____
 _____ + _____ = _____
 _____ + _____ = _____
 _____ + _____ = _____

Pattern 2: Multiplying Fractions

1st Fraction	2nd Fraction	Product
$1/2$	$1/3$	$1/6$
$1/2$	$1/4$	$1/8$
$1/2$	$1/5$	$1/10$
$1/3$	$1/4$	$1/12$

Create Your Own Pattern:

_____ x _____ = _____
 _____ x _____ = _____

$$\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

15. Creating Classroom Activities

Activity 1: Fraction Matching Game

Fraction	Equivalent	Improper	Mixed
$\frac{1}{2}$	$\frac{2}{4}$	$\frac{5}{2} = 2 \frac{1}{2}$	$2 \frac{1}{2}$
$\frac{1}{3}$	$\frac{2}{6}$	$\frac{7}{3} = 2 \frac{1}{3}$	$2 \frac{1}{3}$
$\frac{3}{4}$	$\frac{6}{8}$	$\frac{15}{4} = 3 \frac{3}{4}$	$3 \frac{3}{4}$

Create Your Own Matching Game:

Fractions to Match:

$\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$, $\frac{2}{3}$, $\frac{4}{5}$, $\frac{5}{6}$

Equivalent Fractions:

_____, _____, _____, _____, _____, _____

Activity 2: Create a Fraction Wall

1
 $\frac{1}{2}$ $\frac{1}{2}$
 $\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{3}$
 $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$
 $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$

16. Creation Checklist

When creating fractions

- Choose numerator and denominator
- Is it proper, improper, or unit?
- Can it be simplified?
- Are there equivalent fractions?

When creating problems

- Is the problem clear?
- Are the numbers reasonable?
- Is the operation appropriate?
- Can you solve it?

When creating word problems

- Does it have a real-life context?
- Are the numbers realistic?
- Is the question clear?
- Does the answer make sense?

When creating worksheets

- Are problems varied?
- Is there a mix of difficulty levels?
- Is there space for working?
- Is there an answer key?

When creating BODMAS problems

- Are all operations included?
- Is the order correct?
- Is the answer verifiable?
- Is the problem clear?

When creating equivalent fractions

- Are both numerator and denominator multiplied/divided by the same number?
- Is the value unchanged?
- Is the new fraction in simplest form?

17. Self-Test - Creation Questions

No.	Task	Your Answer
1	Create 3 proper fractions	_____, _____, _____
2	Create 3 improper fractions	_____, _____, _____
3	Create 3 mixed numbers	_____, _____, _____
4	Create 3 equivalent fractions for $\frac{3}{5}$	_____, _____, _____
5	Create an addition problem with different denominators	_____
6	Create a subtraction problem with mixed numbers	_____
7	Create a multiplication problem with cancellation	_____
8	Create a division problem with whole number $\div$ fraction	_____
9	Create a word problem involving fractions	_____
10	Create a BODMAS problem with 3 operations	_____

18. Creation Templates

Template 1: Create from Conditions

Denominator: _____

Numerator: _____

Fraction: _____

Equivalent: _____, _____, _____

Simplified form: _____

Template 2: Create Problem Type

Problem Type: _____

Problem: _____

Solution Steps:

1. _____

2. _____

3. _____

Answer: _____

Template 3: Create Word Problem

Context: _____

Numbers: _____

Question: _____

Solution: _____

Answer: _____

Template 4: Create Equivalent Fractions

Original: _____

x2: _____

x3: _____

x4: _____

x5: _____

Revision Tip: Creation is the highest level of thinking! Be creative, design your own problems, worksheets, and patterns. The best way to master fractions is to create problems for others to solve!